Submitted By: Group 12
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2024S 052411-1 Business Intelligence 1

Assignment 1

Topic & Data Selection

The subject we chose for our assignment is from Retail (sale) Domain.

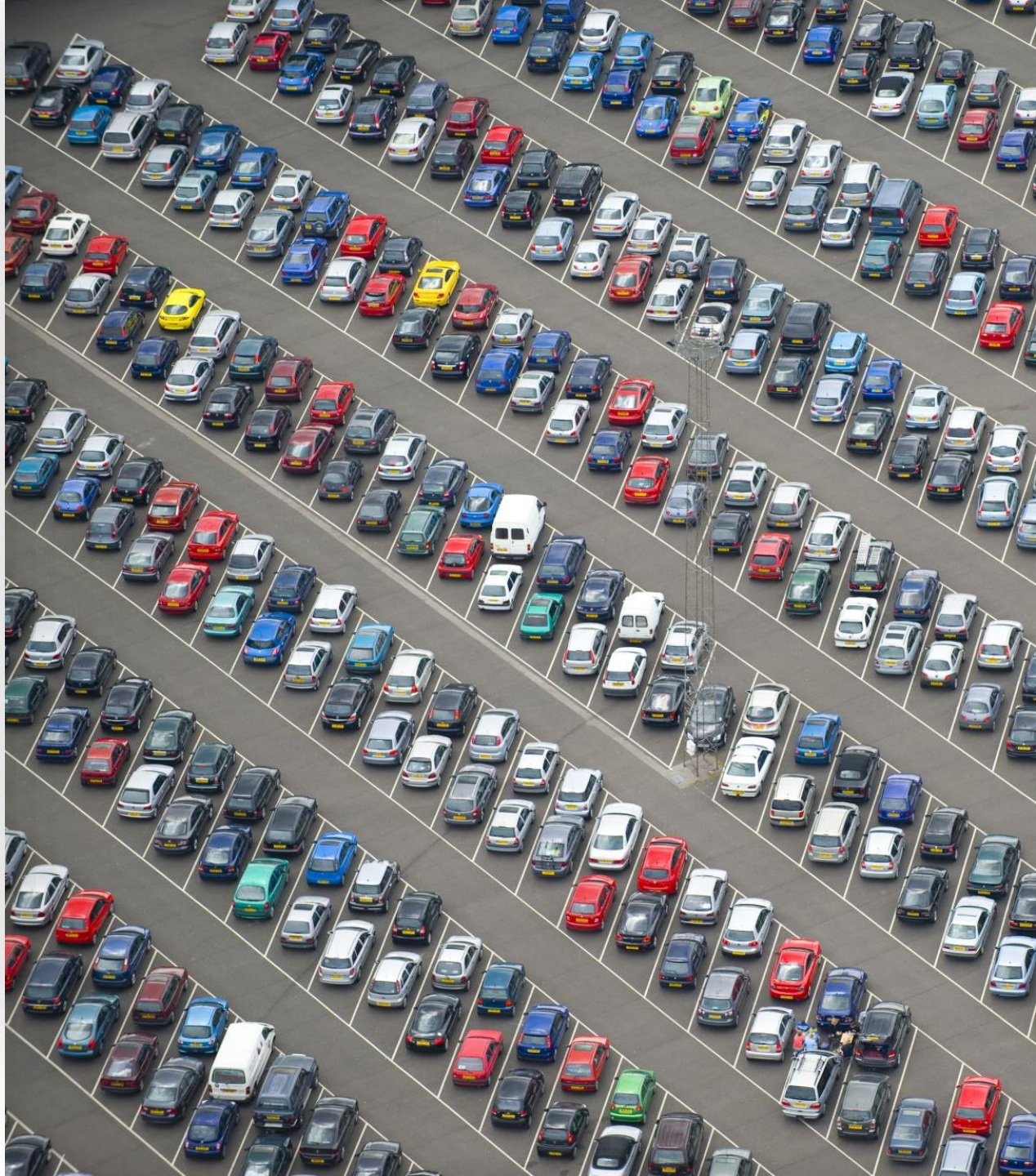
We have chosen two datasets

1. German Car Insights
2. Car Sales

Below are the links attached for both the datasets:

https://www.kaggle.com/datasets/yaminh/german-car-insights

https://www.kaggle.com/datasets/gagandeep16/car-sales



BI Justification & Focus

Different analytical questions that can be derived from these datasets are:

1. **How do Car sales vary across different brands or Models?**

The price variations among different brands and models would guide us about the pricing strategies, dividing cars are into different categories of budget and which are some premium cars, this would also help us analyze customer preferences and response across different brands and their prices.

2. How fuel efficiency relates among different car models?

The higher engine power is the more fuel would be consumed; the more fuel is consumed this means it would lead to more air pollution. Knowing the engine power, it tell us about their fuel consumption which can help customers make choices about their cars.

BI Justification & Focus

3. Which feature of the car has more impact on car sales?

Understanding which feature of the car has higher impact on sales can lead to effective sales strategies, production, prioritizing features that increase sales to increase profit, increase customer satisfaction.

4. How does Engine Power relate to car price and mileage?

These all are interconnected factors that affects the customer choice, purchasing decisions, and the overall value of vehicles. It is crucial to find the right balance between these factors for the manufacturers so they can meet the market demand and can satisfy customer decisions.

5. Which Vehicle Types Are Associated with Higher Resale Values?

Knowing which vehicles have higher resale value has a significant impact on both customers and companies. It would help customers to make smarter purchasing choices and help manufacturers to align their production and predict market trends.

Data Modeling

Dataset 1: German Car Insights

This dataset has a large list of different cars, almost 100,000 entries and 15 columns, each column has different attribute providing detailed information on each vehicle, including its brand, model, color, and registration date. It has quantifiable measures like price, power, and mileage. The dataset allows trend analysis in automotive industry, customer preferences and sales analysis. Some of the attributes of this dataset are as follow:

1. Brand
2. Model
3. Color
4. Registration date
5. Year
6. Price in Euro
7. Power
8. Transmission Type
9. Fuel Consumption
10. Mileage
11. Offer Description
12. Registration Date

Data Modeling

Dataset 2: Car Sales Data

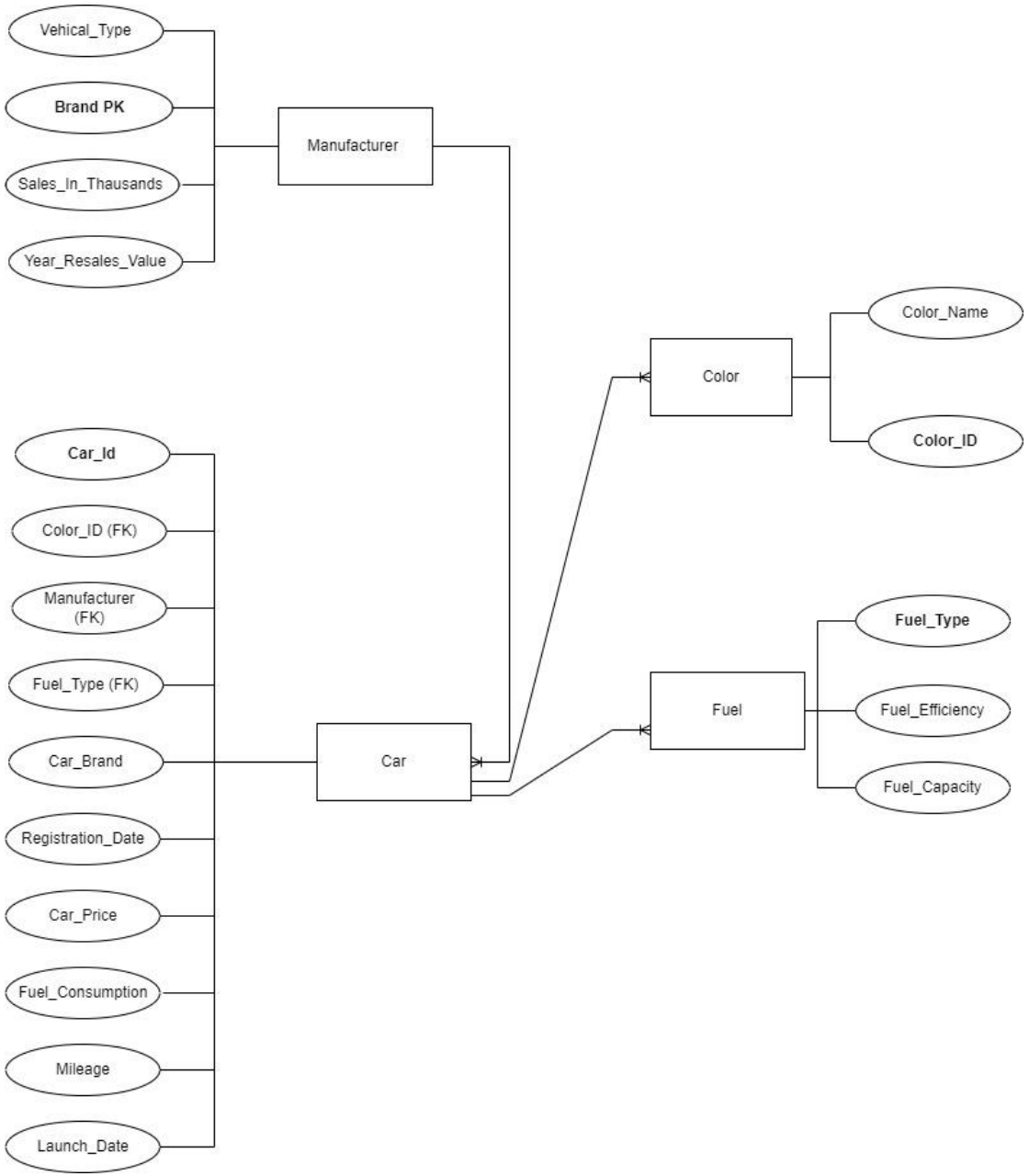
This dataset is about the car sales data and has almost 153 entries with 16 columns and each column describing different attribute. It has quantifiable measures like sale amount, price of cars, engine size. This dataset tells us about the features that has more impact in car sales, characteristics of different vehicles and how can we predict the car sales. Some of the attributes of this dataset are as follow:

1. Manufacturer
2. Model
3. Sales in thousands
4. Year Resale Value
5. Vehicle Type
6. Price
7. Engine Size
8. Horsepower
9. Wheelbase
10. Size (width, length, weight)
11. Fuel (capacity, efficiency)
12. Launch date

Data Modeling

a. Source Data

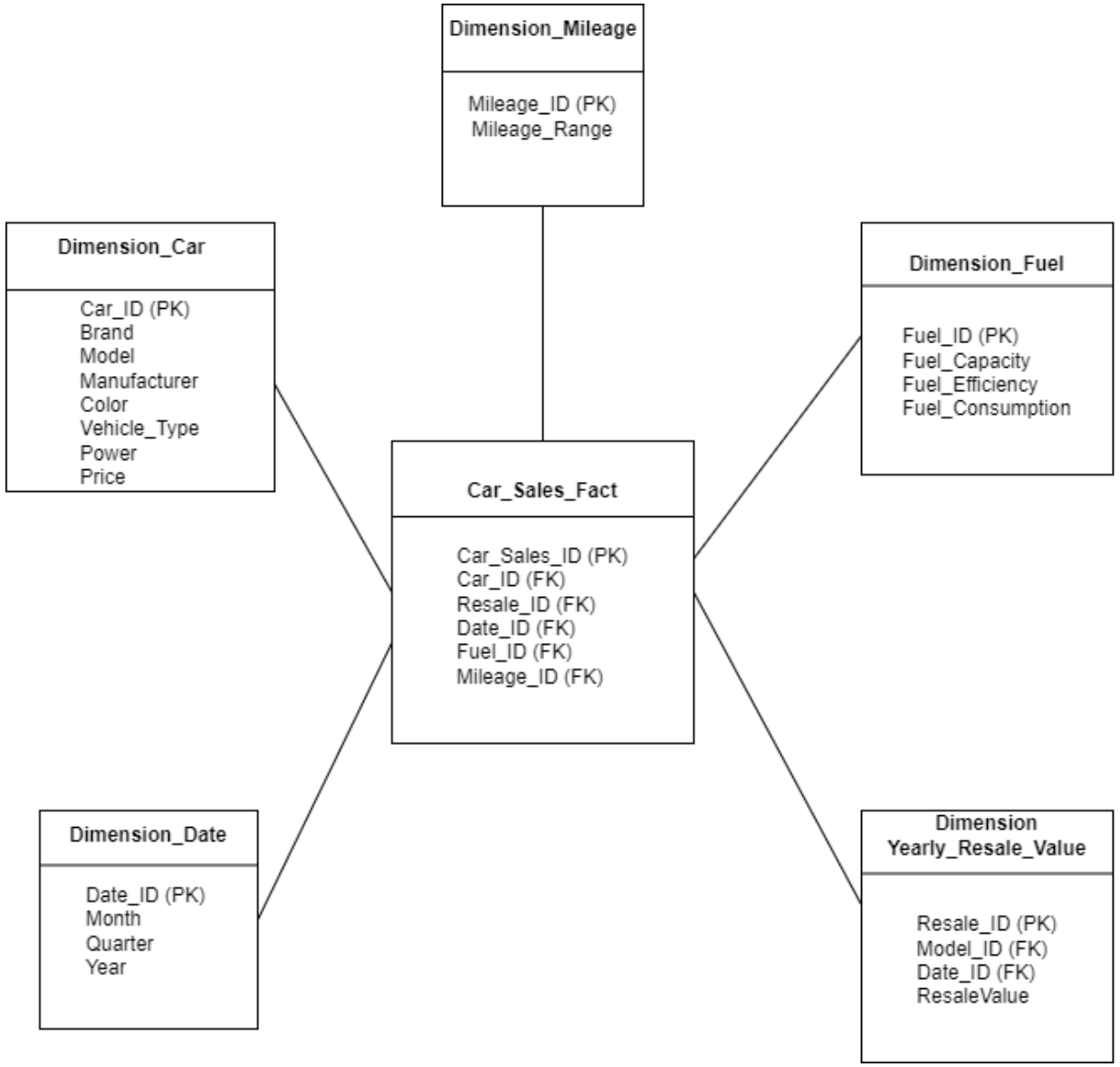
ERD Model



Data Modeling

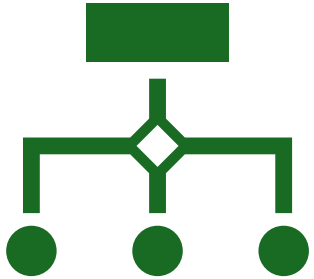
a. BI Data

Star Schema



Data Processing

This part consists of Jupyter notebook file which has been attached with the report. It consist of all the steps taken for data transformation, integration and data cleaning.



This step give us data files which we used for tableau.

Data Processing

a. Data Transformation:

- Download the datasets as csv
- Identified Primary Keys
- Create relationship using Foreign Keys
- Load the dataset into jupyter notebook
- Read the data into pandas data frames
- Displayed first few rows of the datasets to understand the structure
- Identified and discard irrelevant attributes that includes Latest launch, simplified model
- Identified the unnamed: 0 attribute which was irrelevant and unnecessary and removed it

Data Processing

b. Data Integration

- Recreate Dimension_Car from the inner merged dataset with less missing data
- Redo the merge with an inner join to eliminate non-matching entries and minimize missing data
- Create other dimension tables based on the schema

-Dimension_Date

-Dimension_Fuel

- Display unique Manufacturer and Model values from both datasets to understand discrepancies
- Display a sample of unique values from each dataset for comparison simplify model names to match the simpler format of car_sales.csv

Data Processing

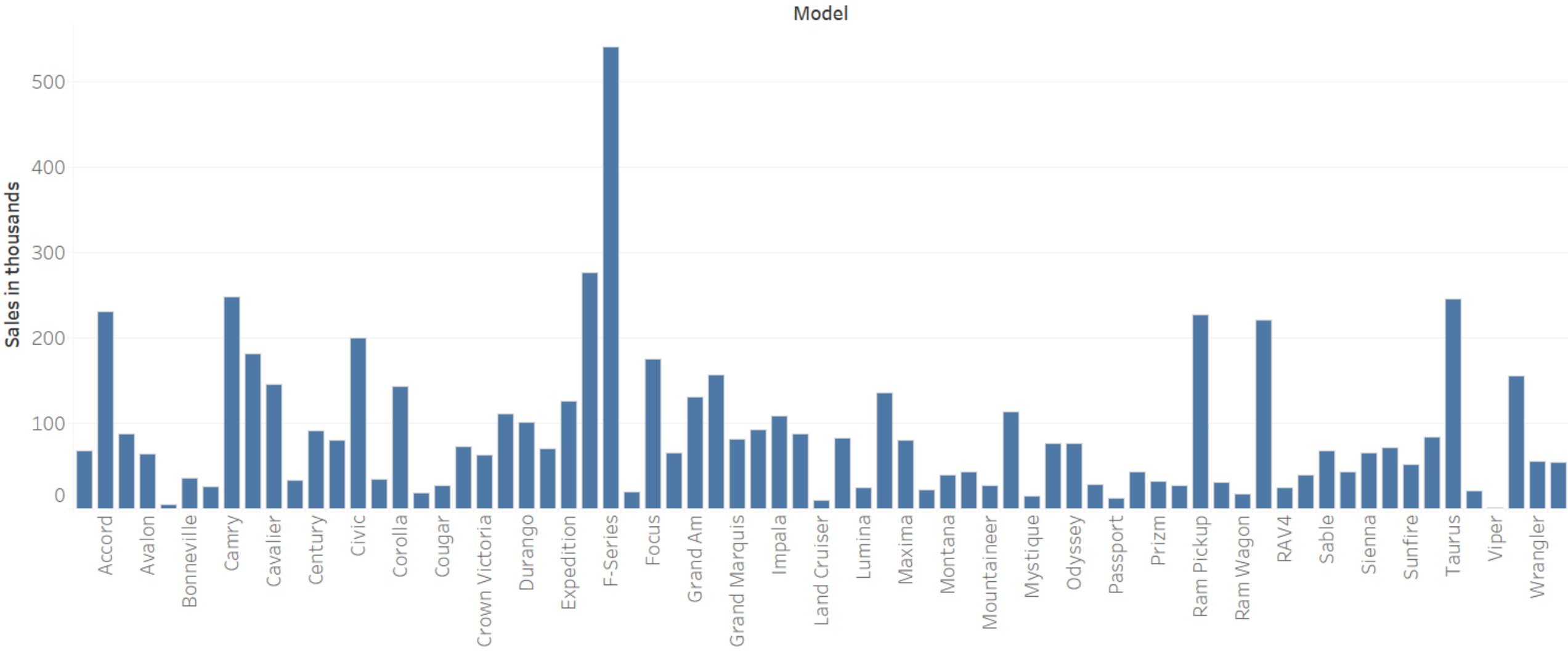
c. Data Cleaning

- Identified missing values
- Fill missing numerical values in 'Car_sales.csv’ using median
- Fill missing numerical values in 'gcar_data.csv’ using median
- Fill missing 'color' in 'gcar_data.csv' with the mode (most common value)
- Fill any new missing values in 'price_in_euro' that resulted from conversion errors with the median
- Convert 'power_kw' and 'power_ps' to numeric values in 'gcar_data.csv' by removing any non-numeric characters and converting
- Convert 'price_in_euro' to numeric, assuming the values might have commas or other non-numeric characters

Data Analytics

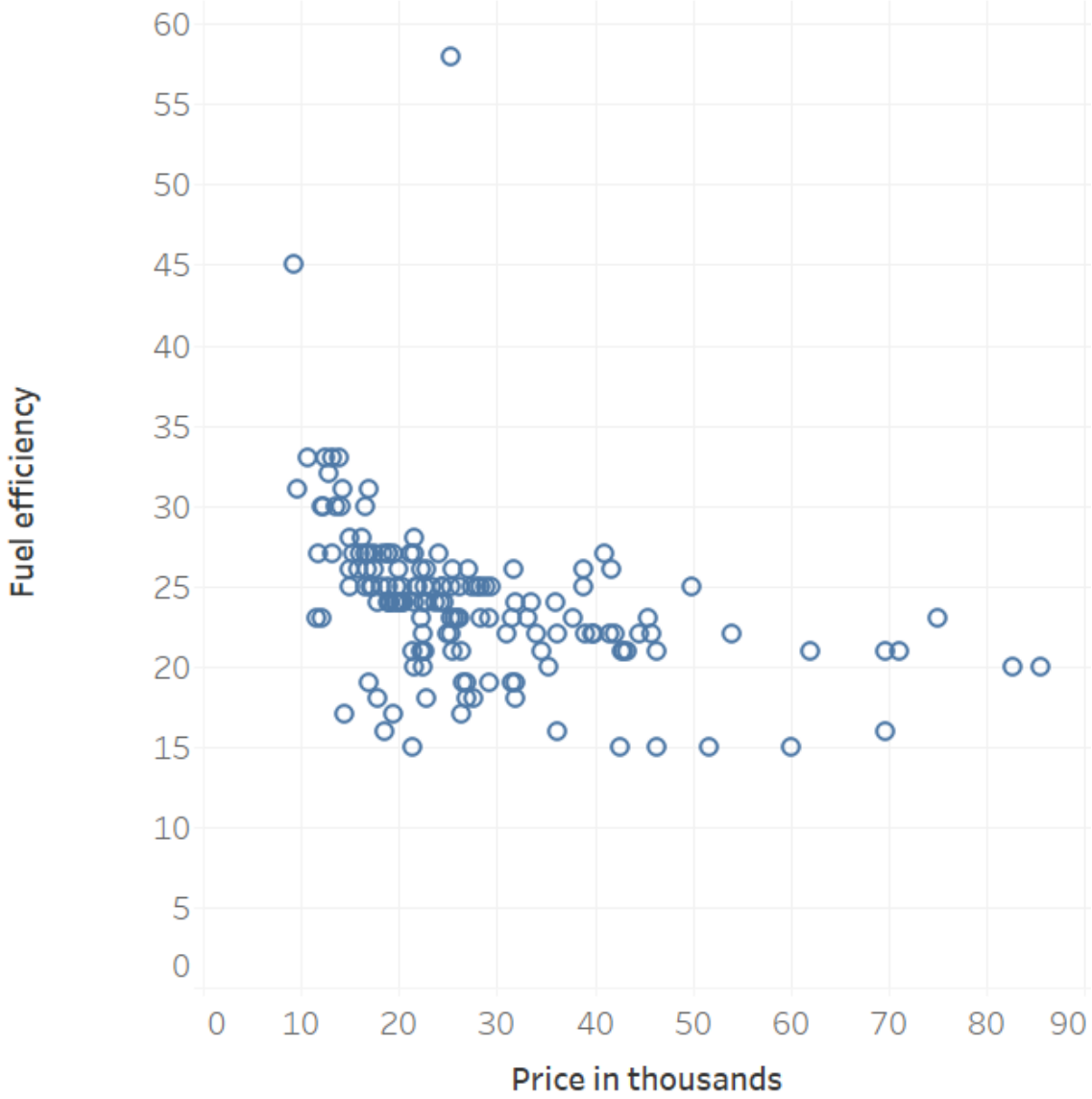
This consist of the part where we loaded our data into tableau and got our insights about the analytical questions. We have attached the screenshot of the insights we got for each question.

Q-1 Car Sales VS Brand



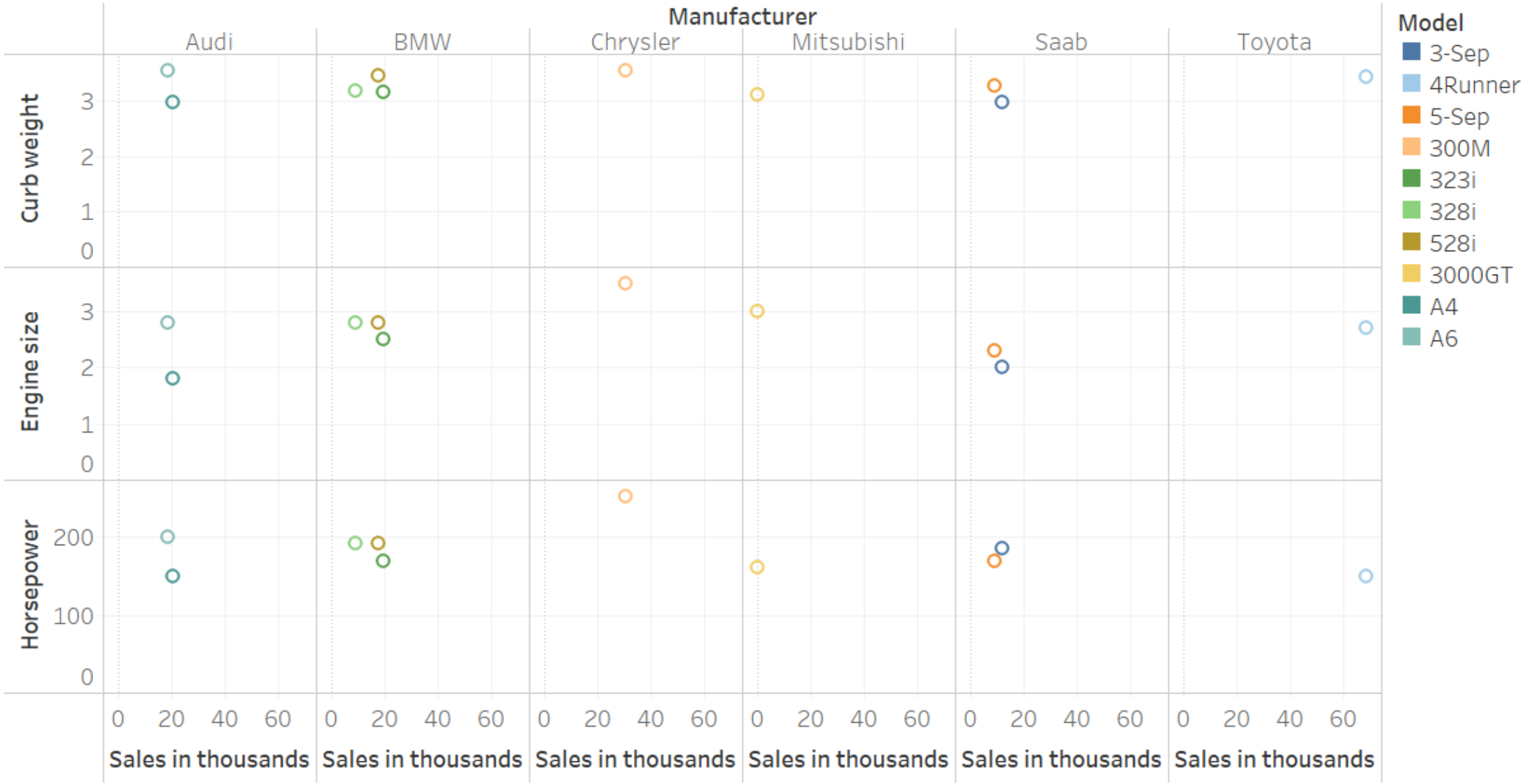
Sum of Sales in thousands for each Model. Details are shown for Model. The data is filtered on Manufacturer, which keeps 10 of 30 members. The view is filtered on Model, which keeps 71 of 156 members.

Q-2 : Relationship Between Car Prices and Fuel Efficiency



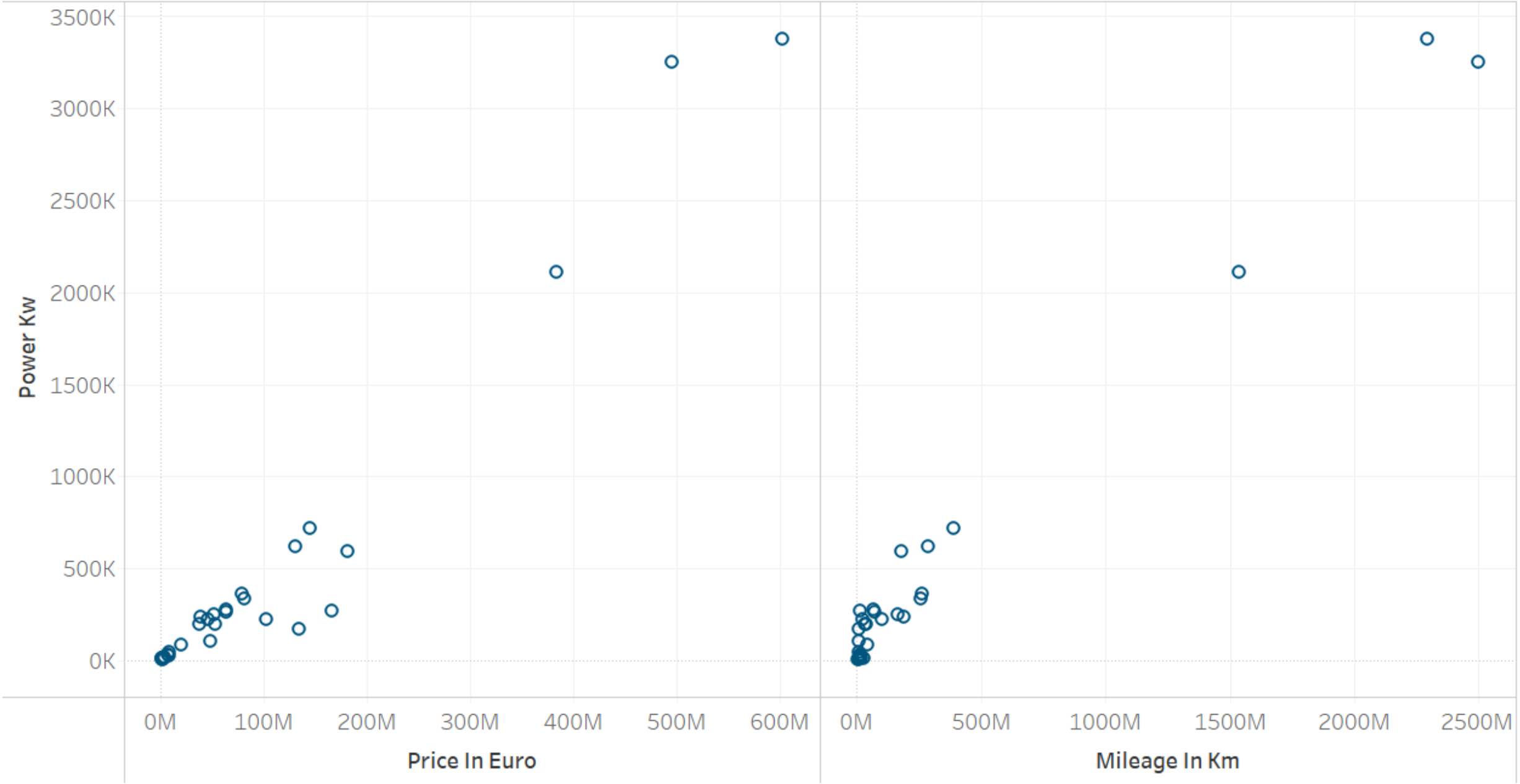
Sum of Price in thousands vs. sum of Fuel efficiency. Details are shown for Model.

Q-3: Which features of car have more impact on car slaes



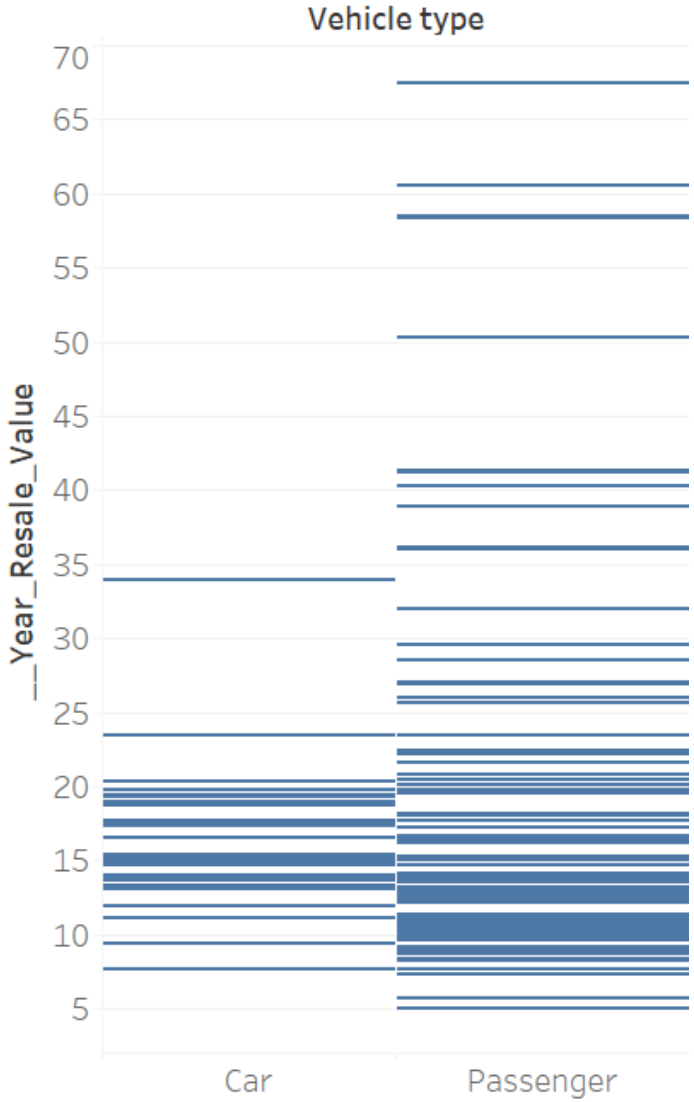
Sum of Sales in thousands vs. sum of Curb weight, sum of Engine size and sum of Horsepower broken down by Manufacturer. Color shows details about Model. The view is filtered on Model and Manufacturer. The Model filter keeps 10 of 156 members. The Manufacturer filter keeps 6 of 30 members.

Q-4: How Does Engine Power Relate to Vehicle Price and Mileage?



Sum of Price In Euro and sum of Mileage In Km vs. sum of Power Kw. Details are shown for Brand.

Q-5: Which Vehicle Types Are Associated with Higher Resale Values?



__Year_Resale_Value for each Vehicle type.
Details are shown for Model.

Reflecting on the Solution:

How good it is?

The solution affectively addresses how the car sales value is affected by different given features in the dataset. It shows necessary insights explaining our analytical questions and showcasing which features tends to increase the sales and which vehicle types has higher resale value. This could be valuable for anticipating market shifts and adjusting business strategies accordingly.

Different stakeholders can use this solution to work on their:

- Manufacturing strategies
- Marketing strategies
- Pricing Strategies
- Inventory Management
- Predicting car sales forecasting.

Reflecting on the Solution:

What else could be done?

Access to a larger dataset could enhance the accuracy and reliability of the results.

Conducting validations and getting feedback from domain expert can help us more enhance our results and can apply in the real-world scenarios.

While the current solution provides valuable insights, there might be chances for further refinement and enhancement. By continuing to iterate and refine the analysis, we can unlock even greater value and drive more informed decision-making.

Work Distribution Slide

Data Research: Both Team Members

Analytical Questions: 2 by Fatima & 3 by Ali

ERD: Fahad Ali

Star Schema: Both Team Members

Jupyter Notebook Work: Both Team Members

Tableau Work: Both Team Members did their own questions

Report: Both Team Members

Difficulties Encountered

Some of the difficulties we faced while working on this data was

- Too many Missing Values
- Data was not organized properly
- Some of the values were in string data type but we need numbers
- Lac of relationship between primary and foreign keys

THANKYOU